AlphaTrace

An evidence-first autonomous financial research investigator

TRACK A

BANKING | NBFCs | FINANCIAL SERVICES

THE IDEA

Given only a NIFTY 50 company, AlphaTrace decides which financial relationships are worth investigating. It turns public filings, peer disclosures and independent sector evidence into one auditable discovery - not a finance chatbot, news summary or investment recommendation.

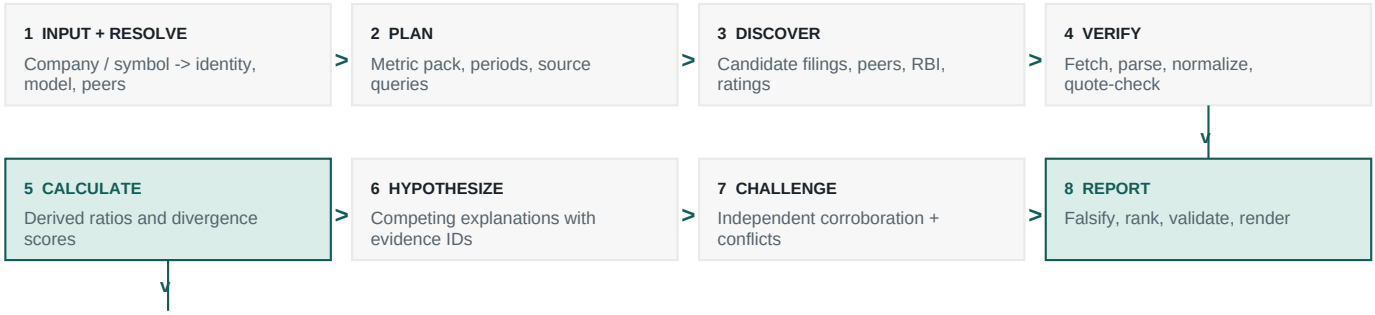
The problem is fragmentation: a company can describe improving asset quality while the flow of stress, its peers, or an external source tells a different story. A single-document summariser cannot reliably expose that relationship.

COMPUTE-FIRST DIFFERENTIATOR

The pipeline computes a time-indexed grid of metrics first, detects a divergence second, and asks the model to explain only validated patterns. The anomaly is found by arithmetic; prose cannot invent the number or URL.

RESEARCH PIPELINE

The same sequential coordinator drives offline fixtures and live public-source runs. Each stage emits inspectable artefacts; rejected hypotheses and missing coverage remain visible.



If evidence is insufficient, return to planning or report: No strong discovery found.

DISCOVERY MECHANISM

The first Track A pack asks whether headline asset-quality improvement is supported by the underlying flow of stress. The lens library is reusable; the resolver selects the applicable pack for Bank, NBFC, Insurer or Holdco.

- Stock vs flow: GNPA with NNPA, write-offs, credit cost or provision coverage.
- Self vs peer: distinguish a company-specific signal from sector normalization.
- Narrative vs number, segment vs aggregate, and entity vs counterparty where data permits.
- Temporal persistence: align periods, detect when the relationship changed, and test base effects.

CONFIRMED NOW / FINALS NEXT

CONFIRMED IN REPOSITORY

- Offline end-to-end research engine and dashboard.
- Deterministic calculations, typed report objects, claim ledger and SQLite replay.
- Public URL validation, bounded retrieval/cache, page-aware parsing and quote verification.
- Conservative live mode that downgrades or withholds weak findings.

FINALS WORK

Broaden peer/source coverage, metric packs and cold unseen-company validation; keep every limitation explicit.

Design rule: deterministic code owns arithmetic, source identity, quote status, independence metadata, conflict classification and final schema validation.

Public sources only | No investment recommendations | Company-specific conclusions are never hard-coded

Make every conclusion contestable

Evidence triangulation, output contract, generalisation and delivery plan

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EVIDENCE + CONTRADICTION

Search results are candidates, not evidence. A source enters the ledger only after public-URL validation, bounded retrieval, parsing and exact or controlled-normalized quote verification.

- Every fact stores metric, value, unit, period, scope, source URL, publisher family, tier, page and exact passage.
- Every claim points to evidence IDs and calculation IDs; displayed numbers come from deterministic Python code.
- Independent-family count prevents three company-authored documents from masquerading as three confirmations.
- Typed conflicts surface narrative/metric, restatement, scope, timing and company/peer mismatches instead of hiding them.

Evidence ladder

T1	NSE/BSE filings, results, annual reports
T2	Investor presentations and earnings calls
T3	RBI / IRDAI / SEBI independent data
T4-T6	Ratings, peers, counterparties, credible media

SELF-FALSIFICATION

The leading hypothesis must earn its rating

T1 **Sector**
Do most peers show the same divergence?

T2 **Accounting**
Could scope, M&A or definition drift explain it?

T3 **Base effect**
Does it survive two- and three-year views?

T4 **Macro / regulatory**
Does timing align with an industry event?

T5 **Disclosure change**
Did the company change what it reports?

PASS / FAIL / INCONCLUSIVE is evidence, not decoration.

ILLUSTRATIVE FINAL REPORT OBJECT

The renderer receives one validated object with all eight required elements. The sketch below shows the shape, not a researched company result or fabricated figure.

1 DISCOVERY
[one clear non-obvious finding]

2 WHY IT MATTERS
[3-5 significance bullets]

3 EVIDENCE
[independent sources + clickable links]

4 TIMELINE
[when the change emerged]

5 NUMBERS
[formula -> inputs -> result]

6 COUNTERARGUMENT
[strongest alternative / conflict]

7 FALSIFICATION
[future condition that changes view]

8 STRENGTH
[Strong / Moderate / Weak + why]

A click on any major claim opens its document, page, passage, period and calculation trace.

GENERALITY

Constants: typed schemas, source policy, quote checks, calculation engine, conflict types, five tests and report contract. Dynamic at runtime: company identity, business model, peers, periods, source plan, metric pack and hypotheses.

A resolver classifies Bank / NBFC / Insurer / Holdco and selects a compatible pack. Missing disclosures reduce evidence strength or stop a conclusion; a new company does not require a new answer template.

TOOLS + DATA

STACK

Python 3.12.8; explicit sequential coordinator; vanilla HTML/CSS/JS; SQLite; standard-library HTTP/HTML parsing; Docker Poppler boundary for PDFs; optional Tavily search and structured OpenAI-compatible provider.

No multi-agent framework is required to make the trail auditable.

BUILD PLAN TO FINALS

18-19 OCT	Freeze Round 1 report; run offline end-to-end checks; review source and limitation wording.
20-21 OCT	Broaden peer retrieval and independent-family scoring; extend bank/NBFC/insurer metric coverage.
22 OCT	Harden temporal alignment, typed contradictions and five falsification tests; preserve weak-result fallback.
23-24 OCT	Cold-run unseen NIFTY 50 and other listed Track A companies; Docker build; cache a recovery run; rehearse demo.
25 OCT	Finals: live judge-selected company, discovery, evidence trail, counterargument and falsification condition.